

Statistical Analysis of Student Learning Preferences at HCT

Research Team

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Course

CDS 3523 - Statistical Inference



Understanding Student Learning Preferences in the Digital Age

Context and Background

The COVID-19 pandemic changed education worldwide. Schools moved to online and hybrid learning models. Now, as campuses reopen, students can choose between three learning modes: in-person, hybrid, or fully online.

Research Question

Is there a statistically significant association between the preferred learning mode (in-person, hybrid, online) and the year of study among computer science students at the Higher Colleges of Technology?

Why This Matters ?

If student preferences change as they progress through their studies, HCT can design better programs for each year level. This helps improve student satisfaction and learning outcomes.



Research Design and Data Collection Approach

53

Student Responses

Survey Instrument

An online survey was designed consisting of **10** relevant questions, structured to include both **categorical** and **continuous** variables. The survey was created using **Google Forms**.

Sampling Method

A **non-probability convenience** sampling method was employed. The online **survey link** Shared through student groups. **All undergraduate students** enrolled in the **Computer Science** division at **HCT**.

Key Variables

Variable	Type
Year of Study	Categorical (Ordinal)
Preferred Learning Mode	Categorical (Nominal)

Data Preparation

- **No** missing data found
- **Combined** 4th Year+ students
- Ready for statistical testing

Normality Testing for 'Satisfaction Rating'

Purpose

We tested if student satisfaction scores follow a normal distribution (bell curve). This is important because many statistical tests require normally distributed data.

Methods Used

- **Histogram:** Graphical representation of data distribution
- **Q-Q Plot:** Compares sample data against theoretical normal distribution
- **Shapiro-Wilk Test:** Formal statistical test providing a p-value

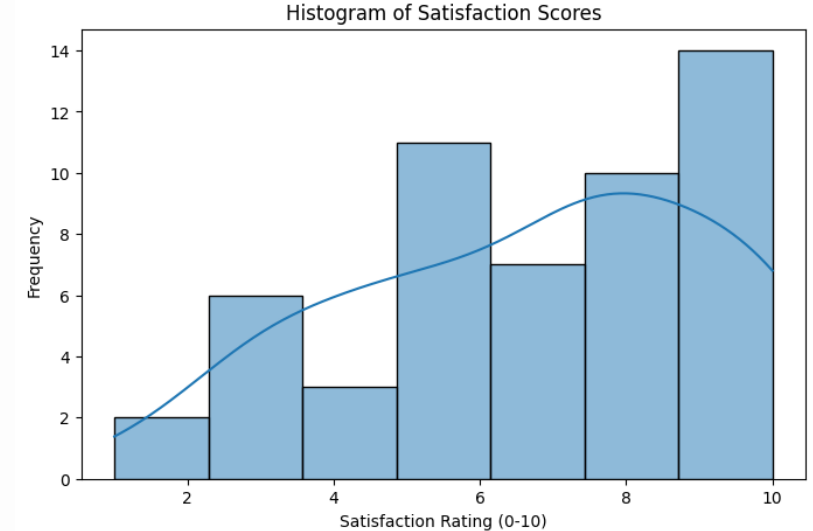
Shapiro-Wilk Test Result

P = 0.004

P-value < 0.05 means **NOT normal**

Conclusion

The satisfaction data does **NOT follow a normal distribution**. We need to use non-parametric tests for further analysis. **(Reject H0)**



Key Finding: Is There a Relationship?

Statistical Test Used

The **Chi-Square (χ^2) Test** of **Independence** was used to assess the association between **Year of Study** and **Preferred Learning Mode**.

Hypotheses

- H₀:** There is **no statistically significant association** between year of study and preferred learning mode.
- H₁:** There is a **statistically significant** association between year of study and preferred learning mode.

Contingency Table - Observed Frequencies

Year of Study	Hybrid	In-person	Online	Total
1st Year	1	1	1	3
2nd Year	3	4	1	8
3rd Year	20	15	1	36
4th Year	2	3	1	6
Total	26	23	4	53

Test Results

Chi-Square (χ^2) Statistic: 4.54
Degrees of Freedom (df): 6
P-value:
0.604

Interpretation

The p-value (0.604) is substantially **greater** than the significance level of 0.05. Therefore, we fail to **reject the null hypothesis (H₀)**.

Conclusion

There is **NO statistically significant relationship** between year of study and learning mode preference.

The differences we see across year levels are likely due to random chance, not a real pattern.

Conclusion & Recommendations

Summary of Key Findings

No relationship between year of study and learning preference

1. **Hybrid** mode most popular (**49.1%**)
2. **In-person** second most popular (**43.4%**)
3. **Online mode** least preferred (**7.5%**)

Student **satisfaction**: 6.74/10 (**room for improvement**)

92%

of students prefer Hybrid or In-person learning.
Students want flexibility combined with face-to-face interaction.

Recommendations

1 Enhance the Hybrid Learning Model

Continue investing in hybrid learning. Train instructors on best practices for blended learning. Ensure technology infrastructure is reliable and easy to use.

2 Conduct Qualitative Research

Use focus groups and interviews to understand WHY students prefer certain modes. Identify specific factors affecting satisfaction to improve curriculum and support services.

Study Limitations

- Sample heavily skewed to 3rd Year students (67.9%)
- Sample size of 53 is relatively small
- Future research needs balanced representation

Future Research

- Use **stratified sampling** for balanced year representation
- Aim for **larger sample size** (100+ students)
- Explore **qualitative reasons** behind preferences

Thank you for your attention. We are ready for questions.